FLEET TEAM DIGEST

737MAX-FTD-34-19010

Issue Title : Head-Up Guidance System (HGS-6000) and FMC Operational Program Software (OPS) U14/U14.1 Reference Glide Slope Angle Display Difference

Airplane Model	ATA	Minor Model(s)
737MAX	3436-00	-7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
	9E991	

Originated Date	Last Revised Date	Created On
04/17/2020	08/28/2026	05/26/2026
Estimated Completion Date	**Next Update Date**	
11/28/2026	11/28/2026	

Revision Description

No updates.

Description

During production flight testing, it was observed that the reference glide slope (REF G/S) angle displayed by the Collins Aerospace HGS Model 6000 for an ILS approach differed from value displayed by the FMC on the LEGS page when using the U14 FMC OPS.

Background

Prior to the implementation of the U14 FMC OPS, the REF G/S angle value displayed by the 737 MAX HGS-6000 for an ILS approach was provided directly from the FMC navigation database (NDB). For U13 FMC OPS and earlier, the REF G/S angle value displayed by the FMC is based upon the selected runway REF G/S angle identified in the FMC NDB. However, the U14 FMC OPS includes a change where the FMC will provide the reference glide path (REF G/P) angle value and not the REF G/S angle for display on the HGS-6000. The REF G/P angle is a real-time value based upon sensed temperature during the airplane descent.

Previous to the U14 FMC OPS, the FMC used the REF G/S angle value for the selected runway from its NDB and displayed it as a static value for the duration of the approach. The U14 FMC OPS now calculates a G/P angle which is dynamic and will be updated by the FMC to reflect sensed standard temperature based upon the FMC's temperature compensation algorithm included in the U14 FMC OPS. This is a new feature of automatic temperature compensation which is discussed in more detail in the attachment to this Fleet Team Digest (FTD) article.

The U14 FMC OPS calculates the REF G/P angle value based on two different points in space and the altitude between these two points will be based on a temperature compensated altitude (i.e. density altitude). The U14 FMC OPS performs this

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compensation for every descent with or without the U14 Automatic Temperature Compensation feature enabled. Using the U14 FMC OPS, the displayed REF G/P angle value compensated for Standard Temperature will typically not be identical to the NDB REF G/S angle. It is possible the REF G/S angle displayed on the HGS-6000 will be different than the REF G/S angle value for the selected approach. Additionally, the FMC REF G/P angle value may change during the course of the approach since the U14 FMC OPS treats this as a dynamic value.

NOTE: For additional details pertaining to the U14 FMC OPS, please refer to 737MAX-FTD-34-18005.

Status

As stated previously, the U14 FMC OPS adds compensation and displays the REF G/P angle based upon standard temperature, rather than the REF G/S angle value from the NDB. The additional compensation to standard temperature may result in a slight change to the REF G/S angle value (for example a 3.00 degree REF G/S angle could become 3.04 after compensation). There may be a perceptible difference noted on the flight path vector displayed on the HGS, which will be less perceptible during the course of the descent to the runway.

Collins Aerospace is in the process of releasing a service information letter revision to note this change in the FMC behavior and to provide information related to the certification status for the HGS-6000 and in conjunction with the U14 FMC OPS. At this time, Boeing understands that Collins Aerospace will only approve the HGS-6000 configuration using dedicated HUD Control Panel (HCP) as a compatible system with the U14 FMC OPS. To address this, Collins plans to revise the HGS-6000 software and GE Aviation plans to release an update to the U14 FMC OPS, U14.1, which will allow the use of HGS-6000 systems on 737 MAX airplanes equipped with the U14.1 FMC OPS. NOTE: Not all 737 MAX airplanes are equipped with an HCP.

At this time, Boeing anticipates the U14.1 FMC OPS to be available in the third quarter of 2021. Collins has indicated they will be releasing a Service Information Letter prior to availability of the HGS-6000 OPS revision, as well. Operators interested in the HGS-6000 software update are encouraged to contact Collins Aerospace directly, as needed.

NOTE: For additional details pertaining to the U14.1 FMC OPS, please refer to 737MAX-FTD-34-20002.

The HGS Model 6000 represents the following STC:

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- Model 6000 – STC ST02522SE

HGS-6000

- Phase 2.1 SW
- Collins OPS: 072-0428-008
- Boeing PN: 3417-FLD-S50-08

STC (Model 6000) has a separate document within the drawing tree hierarchy that lists the approved sensors.

- The Model 6000 STC has the “Sensor Interchangeability Specification” (Collins doc: 946-6849-001)

Once FMC U14.1 is added to those documents (and approved by the FAA), then it means that the FMC U14.1 software can be installed on any HUD Computer classified as a HGS Model 6000.

Interim Action

Boeing plans to install the FMC OPS U13 on 737 MAX aircraft equipped with HGS-6000 systems until the FMC OPS U14.1 and HGS software updates are available. Boeing proposed this as an interim solution pending availability of the U14.1 and HGS-6000 updated software.

Boeing recommends MAX operators to contact Collins Aerospace regarding their certification status of the HGS-6000 installation prior to installation of U14 or U14.1 (when available).

Final Action

Boeing plans to release a Service Bulletin (SB) to install the FMC OPS U14.1 to address the condition discussed herein and recommends operators contact their Collins Aerospace representative for details related to update to the HGS-6000 software. Operators with Head-Up Display (HUD) cannot operate FMC OPS U14.1 until Supplemental Type Certificate (STC) by Collins is approved.

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FMC OPS U14.1, Model 6000 w/ MCDU interface, Phase 2.1 (MDL Rev. U):

- Submitted to FAA on 1/28/2022
- FAA approval is complete
- Collins SIL revision is attached

FMC 2907C2 Hardware Certification & Availability, HUD STC:

- Operators with HUD cannot operate with FMC HW Model 2907C2 until STC by Collins is approved.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

1. 737 MAX - FMC OPS U14.1 - Boeing Service Bulletin Release | 28-May-2022 (Complete)
2. HGS-6000 Phase 2.1 OPS Revision - Certification (for 737 MAX Aircraft Equipped with HGS-6000 and FMC OPS U14 or U14.1) | 21-Jul-2022 (Complete)
3. HGS-6000 HUD STC + C2 FMC Hardware | Q3-2026 (Target)
4. HGS-6000 HUD + C2 FMC Hardware Production Incorporation | Q3-2026 (Target)

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737MAX-FTD-34-18005		
Service Bulletin (SB)	737-34-3656		01/15/2021
Service Bulletin (SB)	737-34-3657		01/15/2021

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Update U14.1 FMC OPS	650630-003/ Disk Set 10-62225-1025	GE Aviation	

Related Categories

- Flight Deck Effects and Maintenance Messages : Issue that occurs frequently and causes a flight deck effect or maintenance message
- Flight Operations : Flight Operations
- Significant Event Report : List of significant in-service or maintenance events reported to Boeing that result in Air turnbacks, ground turnbacks, diversions, rejected takeoffs, in-flight shutdowns, or Significant AOGs. High failure rate of costly/critical systems

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Attachments:

a) HGS DATA PAGE

b) U14 Temperature Compensation

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